CHOP: Mobile Operating Assistant with Constrained High-frequency Optimized Subtask Planning

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Problem

> *Ineffective subtasks, where the plan agent generates unachievable subtasks due to the lack of real-world execution knowledge in VLMs*

Free-form VLM plans also add **inefficient** subtasks that stall off the critical path — the paper blames VLMs' lack of GUI planning experience.

Two failure modes of free-form subtask planning

1. **Ineffective.** "Go to Bob's office" cannot run.
2. **Inefficient.** "Wait for Bob's feedback" stalls.
3. **No GUI experience.** VLMs lack it; tuning is costly.

Evaluation setup

CHOP-En · 30 tasks | CHOP-ZH · 200, 30 run | GPT-4o, all methods | SR ≤ 20 actions | real Android devices

- **All online**; human actions are ground truth.
- **6 of 10 apps** out-of-domain (basis from AITZ).

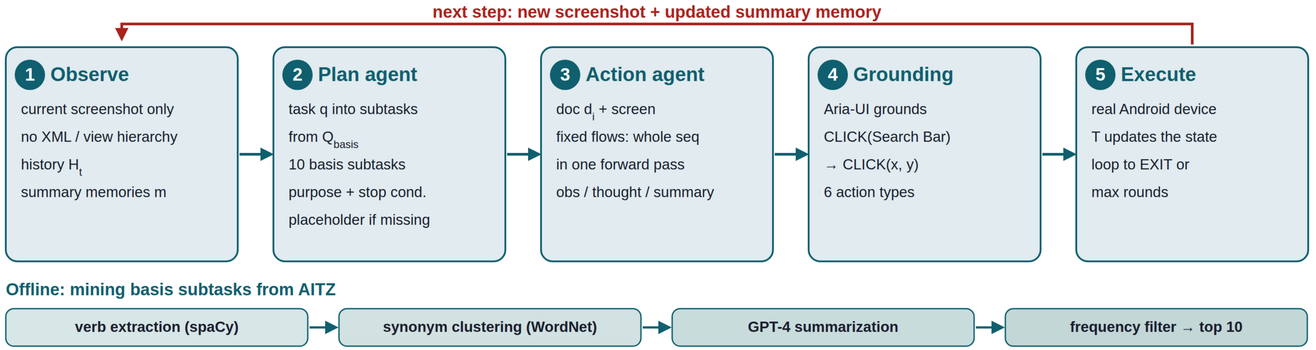
Our read (CUA Radar)

64	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; held-out apps; matched baselines; budget; API cost; ablation; code.
- **Flags:** variance not reported; LLM judge unvalidated; capped rounds; Moba needed Aria-UI to open apps.
- Pattern C17 (manager–worker). Not rerun.

Method: constrain the planner to documented human basis subtasks

- **Sees / acts:** screenshot only (no XML); click, scroll, type, back, exit, wait; Aria-UI maps to x, y.
- **Mechanism:** plan agent must pick subtasks from 10 mined basis subtasks, each with standardized documentation.
- **Efficiency:** fixed-flow subtasks emit the whole action sequence in one call; fewest API calls of all methods.



Our redraw (not the paper's figure): the plan/action loop and the offline mining of basis subtasks.

$$\{q_1, \ldots, q_n\} = f_{\text{plan}}(q,\ Q_{\text{basis}}), \quad q_i \in Q_{\text{basis}}$$

Plan-agent output is constrained to the basis set

Key results: success rate (online, real devices)

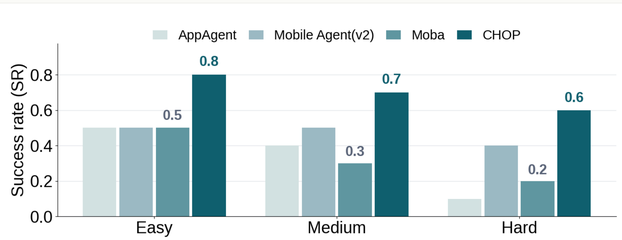


Table 2 (CHOP-En), redrawn: SR within 20 actions; GPT-4o for every method.

Highest SR on CHOP-En at all three difficulties; fewest API calls

Ablation: CHOP-ZH success rate

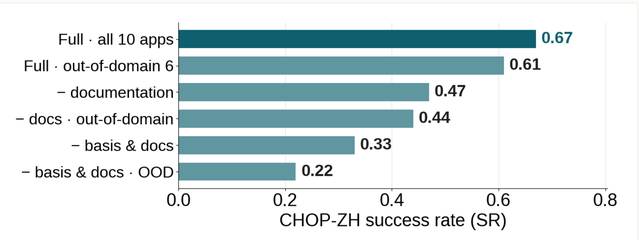


Table 3, redrawn: all 10 apps vs the 6 out-of-domain apps.

SO WHAT → *Docs and basis both matter: out-of-domain SR falls 0.61 → 0.22, in-domain 0.75 → 0.50; CR 0.85 → 0.57 (all apps). Subtask quality (BLEU / ROUGE-L / LLM judge, Figure 3) also improves. CHOP logs 0.0% hallucination, parse and max-iteration errors (Table 4).*

Headline Numbers

0.80
SR, CHOP-En easy
vs 0.50 for AppAgent, Mobile Agent(v2) and Moba (all on GPT-4o)

- **0.70** SR En medium
- **1.00** SR ZH easy
- **0.93** AAC ↓ En hard

The paper's own controls

CONTROL	CHOP	BASELINES
Backbone	GPT-4o, temp 0.0	GPT-4o, all methods
Success budget	SR ≤ 20 actions	same cap
CHOP-ZH hard SR	0.10	Mobile Agent(v2) 0.30
AAC ↓ (En easy)	0.76	Mobile Agent(v2) 3.62
Human baseline	SR 1.00	ground truth

Mobile Agent(v2) wins ZH-hard SR 0.30 vs 0.10

Takeaway

Constrain the planner to 10 documented basis subtasks: better plans, fewest API calls, gains hold on 6 out-of-domain apps.

top-10 basis subtasks | OOD SR 0.61 vs 0.22 | AAC 0.76 lowest | no fine-tuning

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